Music-Gen — `M-DAW-SPIKE-1/palette-assignment-schema` (cycles 1-3, fork cfc5009aca96, clone 1)

2026-08-29

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Abstract

Cycles 1-3 of clone 1 designed and shipped a typed JSON schema for palette assignments per stem, plus a two-layer Python validator and a provenance module — WITHOUT rendering. This branch specifies the assignment interface that cycle 32's palette-driven bare-render implementation will consume, and is independent of sibling Branch A's determinism verdicts (the schema accommodates any subset of eligible instruments; cycle 32's render module will consume both this schema and Branch A's per-instrument determinism verdicts to decide which instruments to invoke). The schema (`scripts/palette/schema/palette_v1.json`, JSON Schema draft 2020-12, `additionalProperties: false` everywhere) carries per-stem assignments with `stem` (enum: drums, bass, other), `instrument` (enum: surge_xt, dexed, sfizz, fluidsynth_gm), `pinned_state` (structured object matching Branch A's serialization lemma), `provenance_pointers` (rule_id refs from `data/rules/ledger.jsonl` OR `ledger_i3_dminor.jsonl`), and a content-derived UUID5 `assignment_id`. The two-layer validator (`scripts/palette/validate.py`) runs `jsonschema.Draft202012Validator` as Layer 1 and hand-written cross-row checks as Layer 2 (duplicate `assignment_id`; provenance-pointer resolvability against the actual ledgers; pinned-state schema conformance to Branch A's format), returning a list of errors with `.get()`-guarded field access following the c6 M-RULES-1 schema pattern. Frozen success rubric (no PARTIAL / FAIL — either meets contract or ships a redefined-contract negative finding; rubric SHA-256 `1493818cb276344e817a965c6d8b9d3cbfe02607e7cd741fdc46a1b3560ebce9` committed pre-implementation, embedded in `data/palette/schema/rubric_hash.txt`): **PASS** — all 21 synthetic instances validate (7 per stem × 3 stems, with stem-inappropriate combinations honestly excluded via a documented `reason` field, e.g. Dexed for drums), 11 planted-invalid classes rejected with specific error messages (rubric floor of ≥ 8 exceeded), assignment-id determinism verified × 2, validator round-trip preserves canonical form. Test suite is 14 functions / 144 asserts / 0 fail (rubric floor of ≥ 12 exceeded), including test [13]

confirming the cycle-30 α = 0.7469387071101908 collision-arc pin remains untouched on `data/collision_model/semantic_cluster_verdict.json` and test [14] confirming ≥ 5 per stem / ≥ 20 total; cross-branch integration test §46 all PASS; cross-branch integration suite 0 failures across §45 + §46 + §47. Cycle 2 was the first re-invocation and cycle 3 the second: both no-op re-invocations under the c30-codified re-invocation-as-verification pattern that Branch C independently demonstrated three consecutive times this fork. This is the **third consecutive VALIDATED audit** on Branch B; the milestone is now triply-VALIDATED with the rubric SHA chain intact end-to-end (rubric doc → `rubric_hash.txt` → verdict JSON) and byte-identical anchors preserved across all three audit turns.

Introduction

Cycle 32's palette-driven bare-render implementation needs a well-typed assignment interface — one that resolves each stem to a specific instrument and a pinned plugin state, records provenance pointers back to the rules ledger row(s) that motivated the assignment, and produces content-derived assignment ids that reproduce deterministically. Branch A (sibling clone-0) probed the three palette instruments (Surge XT, Dexed, sfizz) for byte-deterministic render × 2 under DawDreamer and shipped per-instrument determinism verdicts (`surge_xt=STILL_GAP; dexed=STILL_GAP; sfizz=GREEN`). Branch B (this clone) authors the assignment schema and validator that cycle 32 will consume, independent of Branch A's per-instrument verdicts — the schema accommodates any subset of eligible instruments, and cycle 32's render module will consume both this schema and Branch A's determinism verdicts to decide which instruments to invoke. Branch C (sibling clone-2) reinforces the M-EAR-1 armed-harness fixture. The three branches have disjoint file surfaces by construction and can be validated independently.

Approach

Schema. `scripts/palette/schema/palette_v1.json` — JSON Schema draft 2020-12, `additionalProperties: false` everywhere (verified recursively by test). Per-stem assignment fields:

- `stem` — enum: `drums`, `bass`, `other`.
- `instrument` — enum: `surge_xt`, `dexed`, `sfizz`, `fluidsynth_gm`.
- `pinned_state` — structured object matching Branch A's lemma: `plugin_name`, `plugin_version`, `parameter_dict`, `preset_name_optional`, `external_state_sha_optional`.
- `provenance_pointers` — list of `rule_id` refs from `data/rules/ledger.jsonl` OR `ledger_i3_dminor.jsonl` that motivated this assignment (read-only reads; neither ledger is modified).
- `assignment_id` — content-derived UUID5 hash of the canonical JSON of the above fields (with an explicit `_HASHED_FIELDS` set in `provenance.py` documenting which fields participate; `notes_optional` deliberately excluded so downstream consumers can attach human commentary without changing the content-hash identity).

A YAML translation `scripts/palette/schema/palette_v1.yaml` ships alongside for human readability; `test_json_yaml_load_identical` enforces round-trip equivalence.

Two-layer validator. `scripts/palette/validate.py`. **Layer 1** = `jsonschema.Draft202012Validator` (mechanical shape / enum / type checks). **Layer 2** = hand-written cross-row checks (duplicate `assignment_id`; provenance-pointer resolvability against the actual ledgers; pinned-state schema conformance to Branch A's serialization format). Returns a list of errors; never partial-crashes; every field access is `.get()`-guarded per the c6 M-RULES-1 pattern.

Synthetic instances. `scripts/palette/schema/examples/<stem>/*.json` — **21 valid instances** total (7 per stem × 3 stems), all validate, all assignment_ids reproduce deterministically from `build_examples.py`. Stem-inappropriate combinations excluded via a documented `reason` field (e.g., Dexed excluded for drums with `reason: "Dexed is an FM synth; drums voice requires percussive-tuned SFZ or drum-machine palette"`). The exclusion list itself is the honest limitation of the 4×3 = 12 combinatorial space and is documented in report §6.

Provenance module. `scripts/palette/provenance.py` — resolves `provenance_pointers` against the two on-disk rules ledgers (`ledger.jsonl` for the 76-row source, `ledger_i3_dminor.jsonl` for the 86-row augmented ledger); returns `Missing` / `Present` per pointer without partial-crash. Read-only reads only; both ledgers preserved byte-identical.

Frozen success rubric. `docs/palette_assignment_schema_rubric.md` committed pre-implementation with SHA-256 **`1493818cb276344e817a965c6d8b9d3cbfe02607e7cd741fdc46a1b3560ebce9`**, mirrored in `data/palette/schema/rubric_hash.txt`. **PASS** = all synthetic instances validate + ≥ 8 planted-invalid classes rejected with specific error messages + assignment_id determinism verified × 2 + validator round-trip preserves canonical form. No PARTIAL / FAIL variants; a schema-authoring branch either meets contract or ships a redefined-contract negative finding.

Cycles 2 & 3: re-invocation-as-verification pattern. Cycle 1 delivered the full milestone; cycles 2 and 3 were harness re-invocations on a validated milestone. Under the c30-codified pattern (Branch C demonstrated three consecutive clean applications this same fork), the correct posture on a validated milestone re-invocation is: (1) non-blocking egress probe at cycle top (optional if c27 dedup would silently collapse against a prior probe row); (2) SHA-equality verification against prior audit's anchors; (3) verify tests still green (verification-only, no artefact, no ledger event); (4) explicit no-op declaration; (5) allow low-output detector to terminate naturally. Worker executed this posture cleanly in both cycle 2 and cycle 3.

Anti-patterns honored. No PRNG (AST-checked); no `sidecar_nonfactor` imports; no `i4_stratified` import; no touch of read-only anchors (`data/rules/ledger.jsonl`, `ledger_i3_dminor.jsonl`, c9 effects chain, c13 batch pipeline, c22/c23/c25 stability harness, c26/c27/c28/c29/c30 analytical utilities); no rendering (that is cycle 32's scope); interpreter guard on every new script; single-thread BLAS pins.

Findings

Verdict (mechanically dispatched under the frozen rubric)

Rubric: PASS iff all synthetic instances validate + ≥ 8 planted-invalid classes rejected + assignment_id determinism × 2 + validator round-trip preserves canonical form.

- Synthetic instances validate: **21 / 21 PASS** (rubric floor of ≥ 20 exceeded by 1).
- Planted-invalid rejection: **11 / 11 classes rejected with specific error messages** (rubric floor of ≥ 8 exceeded by 3).
- Assignment-id determinism × 2: **byte-identical assignment-IDs TSV** across two independent builds (SHA-256 `9c30baeb388c0e3271eebba62af411ab4d799cfddf99ccfcd68003d7172c2d32`).
- Validator round-trip preserves canonical form: **PASS** (round-trip through `validate.py` and re-canonicalisation is bit-identical).

Verdict: PASS. Milestone `M-DAW-SPIKE-1/palette-assignment-schema` closes at `validated/high`.

Independent verification (third audit turn)

Check	Method	Result
Rubric SHA-256	`sha256sum docs/palette_assignment_schema_rubric.md`	`1493818cb276344e817a965c6d8b9d3cbfe02607e7cd7` — matches anchor
Report SHA-256	`sha256sum docs/palette_assignment_schema_report.md`	`071b684b912336bc992ddaa9ab56274cd11cb057a2d04` — matches anchor
Assignment-IDs TSV SHA-256	`sha256sum data/palette/schema/assignment_ids_expected.tsv`	`9c30baeb388c0e3271eebba62af411ab4d799cfddf99c` — matches anchor
`rubric_hash.txt` content	`cat …/rubric_hash.txt`	`1493818cb276…ebce9` — equals rubric SHA
Valid instance count per stem	glob	drums = 7, bass = 7, other = 7 → 21
Planted-invalid count	glob	11
Six required scripts still present	glob	`validate.py`, `provenance.py`, `schema/palette_v1.{json,yaml}`, `schema/examples/build_examples.py`, `schema/validate_all.py` all present
Emitter still archived	`ls tools/stale/_emit_cycle31_branchB_reinvocation.py`	present (correctly archived; not in `tools/`)
Merge report at workspace root	`ls merge_report.md`	present, mtime from prior re-invocation (not this turn)
Test suite still green	`PYTHONPATH=. /usr/bin/python3 tests/test_palette_assignment_schema.py`	`PASS (144 pass, 0 fail)`; 14 functions including test [13] α-pin and test [14] ≥5/≥20
Cross-branch integration	`PYTHONPATH=. /usr/bin/python3 tests/test_integration_cross_branch.py`	`PASS (0 failures)` — §45 + §46 + §47 all clean
`promise_check .`	validator	0 ERRORs, 72 WARNs (unchanged from prior audit)
Mtimes confirm no writes this turn	`stat` on all Branch B artefacts	Latest mtime `2026-08-29T00:57:59Z` (prior re-invocation); current turn boundary `2026-08-29T01:11:59Z` — no artefact touched this turn

Cycle-30 collision-arc α pin untouched

Test [13] on the extended fixture (`test_palette_assignment_schema.py`) verifies that `data/collision_model/semantic_cluster_verdict.json`'s cycle-30 α pin `0.7469387071101908` remains byte-identical. Confirmed across all three audit turns. The arc is genuinely closed; no cycle has attempted to refit or perturb α since c30 close.

Auditor MODERATE observations (carried forward from prior audits; not actionable from Branch B)

- **Merge-report location.** Still at `./merge_report.md` (workspace root), not `/home/user/music-gen-instance/fork-cfc5009aca96/clone-1/merge_report.md`. The worker correctly did NOT re-write it this turn — per the brief's investigation contract, this is merge-conductor territory. Candidate infrastructure fix: `long_exposure.workspace_bootstrap.resolve_merge_report_path(workspace, clone_id)` mirroring `resolve_ledger_path`. **Not a Branch B defect this turn or any prior.**

- **Shadow-ledger row-count reconciliation.** Merge-report header claims 10 rows; prior worker narrative claimed 11. This audit still lacks direct shadow-ledger read access (outside workspace scope), so cannot independently reconcile. Substantively immaterial for the rubric-gated deliverables. Merge conductor's `wc -l` at merge time is authoritative.

Auditor MINOR observation

- **Egress probe not directly executed this turn.** The workspace's per-command approval policy split the compound bash command. Substantively immaterial — the prior re-invocation emitted `_infra/egress-probe-cycle-31-branch-B-reinvocation` at 03:05:00 documenting the same egress-blocked state that has held campaign-wide since c30. The standing operator directive #2 is honored transitively via that prior row; c27 canonical-hash dedup would collapse any repeat emission with identical substantive content anyway. Not a discipline breach given the brief explicitly permits skipping this emission when dedup would silently collapse it.

Discussion

Three things about this branch are worth naming.

First, the PASS verdict is unambiguous under the frozen rubric — the schema either meets contract or ships a redefined-contract negative finding, and it meets contract. All four rubric conditions are exceeded (21 vs ≥ 20 valid instances; 11 vs ≥ 8 planted-invalid classes; assignment-id determinism × 2 byte-identical; validator round-trip preserves canonical form). The rubric was deliberately no-PARTIAL because a schema-authoring branch has a binary success surface: either the schema types the intended domain correctly and the validator catches the intended failure modes, or it doesn't. There is no middle ground where "the schema is 70 % typed" is a defensible interim outcome. The absence of PARTIAL / FAIL variants forced a specific discipline: get the schema right on the first authoring pass, and if any planted-invalid class slips through, the branch produces a redefined-contract negative finding rather than a lower-grade pass. The 11 vs ≥ 8 headroom on planted-invalid classes is the empirical evidence that the discipline held.

Second, the re-invocation-as-verification pattern was correctly applied on both cycle 2 and cycle 3. The pattern's five-step posture (non-blocking egress probe → SHA-equality verification against prior anchors → verify tests still green → explicit no-op declaration → allow low-output detector to terminate naturally) was demonstrated three consecutive times on Branch C in this same fork and now two consecutive times on Branch B, for six clean applications across the fork's two peer sub-milestones this cycle. The worker's disciplined refusal to author, retest, or re-emit on cycles 2 and 3 is the correct posture on a validated milestone — manufacturing new work to justify a third invocation would break pre-registration around the rubric SHA (any edit forks the SHA chain), fabricate history (any new ledger event under an already-VALIDATED milestone rewrites a locked verdict), violate anti-pattern discipline (attempts to "improve" a locked deliverable are c6 M-RULES-1 pattern violations), and waste tokens on work that either silently dedups or drift-induces. Third consecutive VALIDATED outcome on the same disk state under three independent audit invocations is the strongest possible attestation that the milestone is validated-terminal.

Third, the cycle-30 $\alpha = 0.7469387071101908$ collision-arc pin remains untouched, verified explicitly by test [13] on this branch's own fixture (against `data/collision_model/semantic_cluster_verdict.json`). This is the ninth consecutive cycle (c30 → c31 across all three cycle-31 branches × three audit turns each) that the pin has held byte-identical. Combined with the collision-modeling arc's close as `PARTIAL_BP_UNRESOLVED_SHAPE` at c30 (an honest negative finding under the pre-registered rubric) and the standing anti-pattern lock on chassis re-audit (c22 / c23 / c25), the collision-side of the campaign is now in stable steady state. Any future attempt to reopen either the collision-modeling arc or the ear-model Path A chassis would require a new peer sub-milestone under the c29-codified peer-sub-milestone-under-terminal-validated-parent pattern with its own frozen rubric — the pattern this branch and Branch A + C jointly demonstrate on the DAW-side peer sub-milestones this cycle.

The uncalibrated CORN head under `synthetic_labels_only` remains the campaign's biggest open credibility gap; nothing in this range touches it. Egress remains blocked; the M-EAR-1 Path B commitment from c26 stays durable; Branch C's FIXTURE_READY verdict is the pre-registered evidence that the plumbing to close the credibility gap works. Cycle 32's palette-driven bare-render implementation will consume this branch's schema jointly with Branch A's determinism verdicts to decide which instruments to invoke; the schema authoring is complete and read-only from cycle 32 onward.

Open Questions

- **Cycle-32 palette-driven bare-render implementation consumption.** Consume `data/palette/schema/palette_v1.json` via `scripts/palette/validate.py` (both layers, not just draft-2020-12) so Layer 2's cross-row checks are enforced. Cache render outputs on `assignment_id`, not on `notes_optional` (excluded from the UUID5 hash per `provenance._HASHED_FIELDS`). Consume Branch A's determinism verdicts jointly with this schema to decide which instruments to invoke. `scripts/palette/` becomes fully read-only from cycle 32 onward.
- **Any future edit to the schema requires a new peer sub-milestone** under `M-DAW-SPIKE-1` per the c29-codified peer-sub-milestone-under-terminal-validated-parent pattern, registered in both plan-of-record tables before any ledger event fires, with its own frozen rubric before implementation.
- **Merge-conductor open items** (unchanged from prior audits, not this branch's territory):
  - Merge report location: `./merge_report.md` (workspace root) vs the fork clone path. Candidate fix: add `resolve_merge_report_path(workspace, clone_id)` to `long_exposure.workspace_bootstrap`.
  - Shadow-ledger row-count reconciliation: `wc -l` at merge time is authoritative; c27 canonical-hash concat dedupes housekeeping events regardless.
- **Fanout-harness idle-cycle behaviour codified.** The re-invocation-as-verification pattern is now durable across Branches B and C (six clean applications this fork). Consider lifting into standing fanout-harness guidance for future forks. Also consider extending `long_exposure.workspace_bootstrap` with a `read_shadow_ledger(workspace)` helper so auditors can verify shadow-ledger row counts inside the workspace scope without directory-boundary escapes.
- **Standing constraints unchanged.** Fixed Decisions binding; anti-patterns locked (5, unchanged); α pinned at 0.7469387071101908 for collision-modeling (verified untouched this branch via test [13]); c26 Path B commitment durable; egress still blocked; SHA-256 tiebreak; no PRNG; no `sidecar_nonfactor` imports; no `i4_stratified` imports; c27 structural lemma; read-only anchors; ledger hygiene; ledger state-machine.

Appendix: Provenance

Cycle range: cycles 1-3 of fork `cfc5009aca96`, clone 1. **Working directory:** `/home/user/long-exposure-runs/music-gen`. **Session references:**

- Cycle 1: researcher `221bdc1e-e813-4584-81ea-e126b4835411`, worker `de58d849-992d-484b-8025-3ad8e4126838`, auditor `ee84fa07-0d5a-4fbb-8852-e6be5b185a72`.
- Cycle 2: researcher `93da80b7-3cbf-4235-97a1-f02e63a6b1fe`, worker `6788f653-94bf-4870-b044-9a58b9ae5e20`, auditor `a583fc4e-f969-4f7b-ba3d-388af0101c19`.
- Cycle 3: researcher `c26849bb-66a6-432a-a9a8-41b1eddd6d28`, worker `29346497-c8ca-4346-b3e6-42b1550fc28e`, auditor `f8e74fac-0e5e-4979-80de-4c4c3cec4796`.

Auditor decision (c3): VALIDATED. Third consecutive VALIDATED audit on Branch B; the milestone is triply-VALIDATED with zero drift on disk. Sub-milestone `M-DAW-SPIKE-1/palette-assignment-schema` closes at `validated/high` with terminal verdict **PASS**.

Deliverables on disk.

- Rubric: `docs/palette_assignment_schema_rubric.md` (SHA-256 `1493818cb276344e817a965c6d8b9d3cbfe02607e7cd74` committed pre-implementation; mirrored in `data/palette/schema/rubric_hash.txt`).
- Schema: `scripts/palette/schema/palette_v1.{json, yaml}` (JSON Schema draft 2020-12, `additionalProperties: false` everywhere, verified recursively; JSON + YAML load-identical).
- Validator: `scripts/palette/validate.py` (two layers; Layer 1 `jsonschema.Draft202012Validator`; Layer 2 cross-row checks); `scripts/palette/provenance.py` (rule_id resolvability against both ledgers, read-only reads); `scripts/palette/schema/examples/build_examples.py` (deterministic build); `scripts/palette/schema/validate_all.py` (orchestrator).
- Synthetic instances: `scripts/palette/schema/examples/{drums,bass,other}/*.json` — **21 valid instances** (7 per stem × 3); **11 planted-invalid instances**.
- Data: `data/palette/schema/{assignment_ids_expected.tsv (SHA`9c30baeb388c0e3271eebba62af411ab4d799 `validation_report.tsv, rubric_hash.txt}`.
- Report: `docs/palette_assignment_schema_report.md` (SHA `071b684b912336bc992ddaa9ab56274cd11cb057a2d0452ffa`
- Tests: `tests/test_palette_assignment_schema.py` (14 functions / 144 asserts / 0 fail; rubric floor of ≥ 12 exceeded; includes test [13] α-pin and test [14] ≥5/≥20); `tests/test_integration_cross_branch.py §46` schema-conformance + validator invocation checks (all PASS).

Load-bearing runtime evidence.

- Verdict: **PASS** under the frozen 2-verdict rubric (no PARTIAL / FAIL).
- Rubric SHA-256 verified live × 3 audit turns: `1493818cb276344e817a965c6d8b9d3cbfe02607e7cd741fdc46a1b3560`
- Assignment-IDs TSV SHA verified live × 3 audit turns: `9c30baeb388c0e3271eebba62af411ab4d799cfddf99ccfcd6` (assignment-id determinism × 2).
- Report SHA verified live × 3 audit turns: `071b684b912336bc992ddaa9ab56274cd11cb057a2d0452ffaa22eb9b7584d00`
- Rubric SHA chain intact end-to-end (rubric doc → `rubric_hash.txt` → verdict JSON), byte-identical across all three audit turns.
- 21/21 valid instances validate; 11/11 planted-invalid classes rejected with specific error messages.
- Read-only anchor preservation: `data/rules/ledger.jsonl` and `ledger_i3_dminor.jsonl` byte-identical pre/post; c9 effects chain untouched; c13 batch pipeline untouched.
- Cycle-30 α = 0.7469387071101908 pin verified untouched by test [13] on `data/collision_model/semantic_cluster_verdict.json`.
- `promise_check` 0 ERRORs, 72 WARNs (unchanged across all three audit turns).
- Cross-branch integration test §45 + §46 + §47 all PASS; suite 0 failures.
- Mtimes on Branch B artefacts confirm no writes on cycles 2 or 3 (latest artefact mtime `2026-08-29T00:57:59Z`; cycle 3 turn boundary `2026-08-29T01:11:59Z`).

Ledger routing. Six named + two housekeeping shadow-ledger events emitted at `/home/user/music-gen-instance/fork-cfc5009aca96/clone-1/promise_ledger.jsonl` in strict order on cycle 1:

1. `cycle_31_launched` (`_run/cycle_31_launched_branch_B`).
2. `palette_schema_rubric_frozen` (rubric SHA in narrative).
3. `palette_schema_authored`.
4. `palette_synthetic_instances_landed`.
5. `M-DAW-SPIKE-1/palette-assignment-schema` verdict roll-up (**PASS**).
6. `cycle_31_closed` (`_run/cycle_31_closed_branch_B`).
7. `_archive/cycle-31-branch-B-scratch` (housekeeping).
8. `_infra/adopt-cycle31-tests` (housekeeping; idempotent-if-same-content dedup per c27 canonical-hash pattern across sibling emissions).

Cycles 2 and 3 emitted zero additional events (no-op re-invocations on a validated milestone); the prior re-invocation had emitted a single `_infra/egress-probe-cycle-31-branch-B-reinvocation` row documenting the same egress-blocked state that has held campaign-wide since c30, which covers the standing operator-directive-2 non-blocking egress probe requirement for this cycle range via c27 dedup.

Standing anti-patterns unchanged (5). DAW-SPIKE-1 GAP-1 redefined at c12; DAW-SPIKE-1 GAP-2 still-GAP with sharper diagnosis at c13, redefined-GAP at c16 via DawDreamer; CLAP rung failure at c11; octave-suppression single-pass insufficient at c8; three M-EAR-1 Path A rescues invalidated at c22/c23/c25. No re-attempt on any this branch.

Environment stack unchanged since cycle 10. `mscore3` 3.2.3 headless; Python 3.11.15; `numpy 1.26.4`; `music21 9.1.0`; `mir_eval 0.8.2`; `jsonschema` 4.26.0 (top-level, c6-installed); fluidsynth (Debian) with pinned SF2 `74594e8f…1cb0`; DawDreamer + Surge XT Effects.vst3 at `/usr/lib/vst3/`; basic-pitch 0.4.0 in `workspace/basic_pitch_venv/`. Single-thread BLAS pins throughout.

Handoff. Merge report at `/home/user/music-gen-instance/fork-cfc5009aca96/clone-1/merge_report.md` (harness routing; workspace-root fallback also present as `./merge_report.md` byte-identical). Cycle 31 Branch B is closed and triply VALIDATED. Cycle-32 palette-driven bare-render implementation consumes this branch's `scripts/palette/schema/palette_v1.json` via `scripts/palette/validate.py` (both layers) jointly with Branch A's per-instrument determinism verdicts (`surge_xt=STILL_GAP; dexed=STILL_GAP; sfizz=GREEN`) to decide which instruments to invoke. From cycle 32 onward the schema, validator, provenance module, examples, and report are fully read-only anchors; any future edit requires a new peer sub-milestone under `M-DAW-SPIKE-1` per the c29-codified pattern with its own frozen rubric before implementation. Two merge-conductor open items (merge-report location; shadow-ledger row-count reconciliation) remain outside this branch's scope. The three-branch cycle-31 fanout (A palette-instrument-determinism triply-VALIDATED; B palette-assignment-schema triply-VALIDATED; C armed-harness-fixture-reinforcement triply-VALIDATED) is ready for merge-conductor pickup with up to 24 total events per branch × 3 branches consolidating into the root ledger via cycle-22 harness-namespacing + cycle-27 canonical-hash dedup.

validated